

Breathing Pattern and Rate Prediction Using Wi-Fi Channel State Information

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Abstract—Traditional respiratory monitoring methods rely heavily on contact-based devices that can induce patient discomfort and alter natural breathing baselines. Moving away from computationally intensive traditional model-based processing, this work models respiratory estimation as a continuous regression task. The proposed end-to-end framework extracts raw complex CSI amplitudes and processes them through variance filtering, Z-score normalization, Principal Component Analysis (PCA) dimensionality reduction, and temporal windowing. A hybrid deep learning architecture—combining a 1-Dimensional Convolutional Neural Network (1D-CNN) for local high-frequency feature extraction and a Bidirectional Long Short-Term Memory (BiLSTM) network for capturing bidirectional temporal mechanics—is deployed to map multi-path CSI perturbations directly to Breaths Per Minute (BPM). The network was trained and evaluated using an open-source National Institute of Standards and Technology (NIST) dataset generated by a calibrated RespiPro breathing manikin. Empirical results demonstrate strong predictive performance, achieving a Mean Absolute Error (MAE) of 2.0564 BPM and a Coefficient of Determination (R^2) of 0.8195. Furthermore, 90.73% of the model's predictions fall within a clinically acceptable ± 5 BPM threshold. By transitioning from discrete integer classification to continuous regression, this system allows for the interpolation of fractional breathing rates, paving the way for scalable, non-invasive vital sign monitoring in both clinical and residential environments.

Index Terms—Contactless Respiratory Monitoring, Wi-Fi Channel State Information (CSI), Deep Learning, CNN-BiLSTM

I. INTRODUCTION

Current observation of the movement of breathing reveals key diagnostic information on a range of health disorders and other conditions. Most respiratory monitoring methods rely on some form of contact between the patient and monitoring equipment. When a patient's body is physically attached to a device, it can cause discomfort to the patient. In the process, it can alter the patient's natural breathing baseline, leading to inaccurate respiratory data. To overcome these disadvantages, contactless monitoring systems have come forth as a non-invasive alternative. Although various optical and audio systems have been proposed, their performance is often degraded due to line-of-sight requirements, lighting and background noise [1].

Radio frequency (RF)-based sensing can overcome these environmental constraints by using the emission of electromagnetic waves. In particular, systems that employ Commercial Off-The-Shelf (COTS) Wi-Fi hardware provide continuous

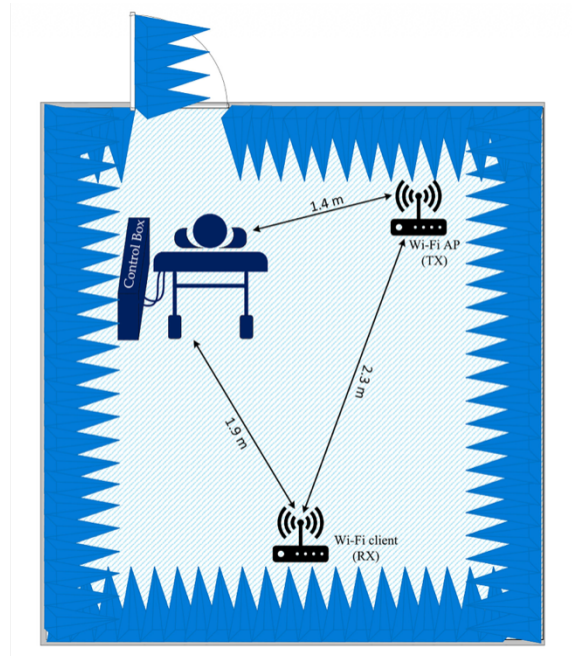


Fig. 1. shows the layout of the measurement setup [1]

monitoring at a low cost and with a high degree of scalability and accessibility without the need for specialist radar equipment. While Wi-Fi signals are transmitted from a transmitter to a receiver, tiny physical changes such as the expansion and contraction of the chest of a human body during the breathing cycle result in modification of signal amplitude and phase. The Wi-Fi Channel State Information (CSI) refers to a fine-grained metric that records the amplitude attenuation and phase shift of the signal across a large number of sub-carriers. These multi-path disturbances can be captured within CSI. By observing these variations in the complex CSI data streams, it is possible to effectively isolate and track respiratory patterns [1].

Traditional model-based processing techniques for CSI sensing typically involve intensive computation which may render them unsuitable for either real-time or resource-limited scenarios. This report proposes a data-driven solution by considering respiratory monitoring as a multi-class sequence classification task. This project employs a deep learning architecture that aims to approximate the complex and nonlinear

mapping between variations in raw CSI data and discrete respiratory rates based on the BreatheSmart Algorithm [1]. Using neural networks substantially lowers the cost of estimating during deployment.

This implementation uses a rigorously controlled and open-source dataset from the National Institute of Standards and Technology (NIST) [2] to train and evaluate the proposed architecture. The raw CSI tensors used in the dataset were collected from a manikin (RespiPro) used to simulate breathing (see Fig. 1). This manikin was pre-programmed to exhibit a variety of clinical breathing rates and patterns (see Table I); this provided the basis to evaluate the algorithm [2]. This report will provide the data preprocessing pipelines, the configuration of the deep learning model, and the empirical results for classifying respiratory motion on this dataset [2].

TABLE I
BREATHING PATTERNS USED IN THE EXPERIMENTS.

Pattern	BPM	Max Inspiratory/Expiratory TV (mL)
#1	25	319
#2	6	705
#3	18	488
#4	18	171
#5	22	651
#6	15	587
#7	28	442
#8	19	513
#9	25	463

II. METHODOLOGY

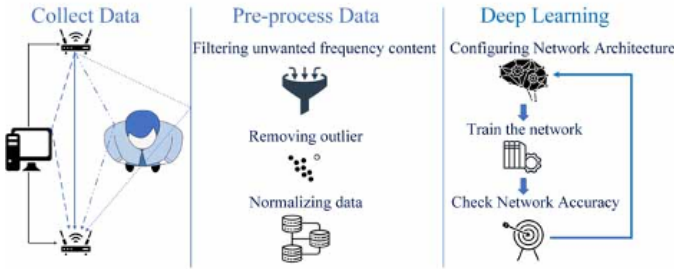


Fig. 2. End-to-end architecture of the proposed system.

As shown in Fig 2, the end-to-end architecture of the proposed contactless respiratory monitoring system. The pipeline is divided into two primary stages: a data processing sequence that extracts and normalizes the relevant frequency-domain features from the raw Wi-Fi signals, and a hybrid deep learning architecture that maps these spatio-temporal features to a continuous respiratory rate prediction.

A. Data Processing Pipeline

The raw dataset consists of Wi-Fi Channel State Information (CSI) collected in a controlled environment. To prepare this data for the neural network, an automated pipeline (see Fig. 3) processes the real and imaginary components of the CSI streams through the following steps:

- **Amplitude Extraction:** The pipeline recursively scans the dataset directories to load the experimental configurations and their corresponding CSI logs. The raw CSI amplitude is calculated from the real and imaginary tensors using the standard magnitude formula:

$$X = \sqrt{\text{real}^2 + \text{imag}^2}$$

- **Filtering and Normalization:** To eliminate dead channels or static environmental reflections, sub-carriers with near-zero variance (standard deviation $\leq 10^{-6}$) are removed. The remaining dynamic streams undergo Z-score normalization, scaling each feature to a mean of zero and a standard deviation of one to stabilize gradient descent.
- **Dimensionality Reduction (PCA):** To manage computational complexity and extract the most significant variance associated with human respiration, Principal Component Analysis (PCA) is applied to the normalized amplitude matrix. The data is reduced to a maximum of 20 principal components. If an observation contains fewer than 20 valid features, zero-padding is applied to ensure uniform tensor shapes across the dataset.
- **Temporal Windowing:** The continuous time-series data is segmented into fixed-length sequential inputs compatible with recurrent neural network layers. A sliding window algorithm is applied with a window size of 100 timesteps and a step of 15.
- **Dataset Splitting:** The aggregated observation windows and their corresponding ground-truth Breaths Per Minute (BPM) labels are stacked and split into an 80% training set and a 20% hold-out testing set.



Fig. 3. Datapipeline

B. Hybrid CNN-BiLSTM Architecture

The predictive model treats the estimation of the respiratory rate as a continuous regression problem. To effectively capture both local signal variations and long-term breathing cycles, the

model utilizes a sequential deep hybrid Convolutional Neural Network and Bidirectional Long Short-Term Memory (CNN-BiLSTM) architecture as shown in Fig. 4.

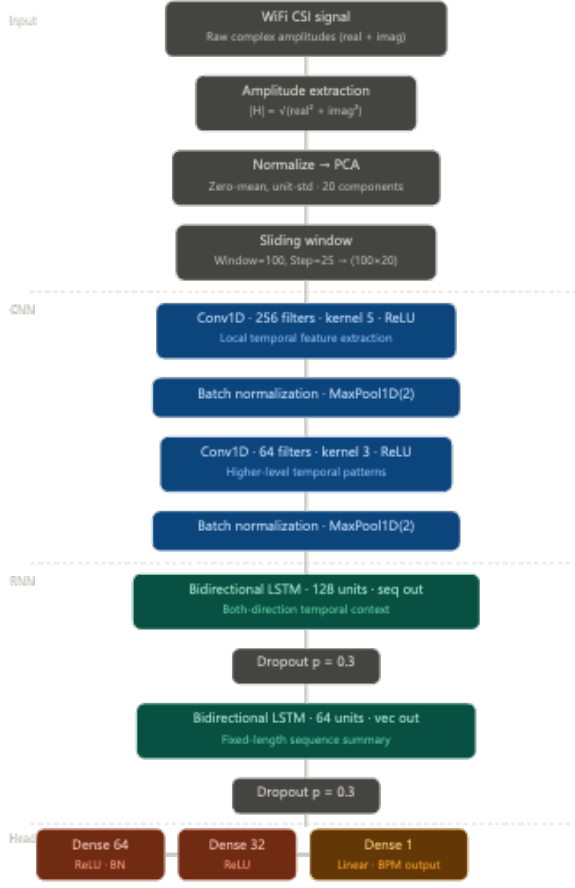


Fig. 4. Model architecture of respiratory monitoring system

The network is structurally divided into three functional blocks:

1) **Spatial Feature Extraction (CNN Block)**: The pre-processed sequential CSI features are first passed through a one-dimensional Convolutional Neural Network (CNN) block designed to extract local temporal patterns and high-frequency dependencies from the signal.

- **Conv1D Layer 1**: Utilizes 256 filters with a kernel size of 5 and a Rectified Linear Unit (ReLU) activation function.
- **Batch Normalization and Pooling**: The output is batch-normalized to stabilize the learning process, followed by a MaxPooling1D layer (pool size = 2) to down-sample the feature maps.
- **Conv1D Layer 2**: A second convolutional layer employs 64 filters with a kernel size of 3 and a ReLU activation function.
- **Batch Normalization and Pooling**: A second Batch Normalization layer followed by MaxPooling1D (pool size = 2) further reduces the feature dimensions before recurrent processing.

2) **Sequential Learning (RNN Block)**: The spatially reduced feature maps are then passed to a recurrent neural network (RNN) block to capture long-term bidirectional temporal dependencies within the breathing cycle.

- **BiLSTM Layer 1**: A Bidirectional Long Short-Term Memory (BiLSTM) layer with 128 hidden units processes the sequence in both forward and backward directions while returning the full sequence. A Dropout layer (dropout rate = 0.3) follows to reduce overfitting.
- **BiLSTM Layer 2**: A second BiLSTM layer with 64 hidden units processes the sequence and returns only the final hidden state, producing a compact high-level feature vector. A second Dropout layer (dropout rate = 0.3) is subsequently applied.

3) **Fully Connected Regression (Dense Block)**: The extracted spatio-temporal features are passed through fully connected layers to predict the continuous respiratory rate.

- **Hidden Dense Layers**: A dense layer containing 64 neurons with ReLU activation is followed by Batch Normalization and a second dense layer with 32 neurons using ReLU activation.
- **Output Layer**: The final dense layer consists of a single neuron with a linear activation function, producing the predicted Breaths Per Minute (BPM).

C. Training and Optimization Configuration

The model was compiled and trained using robust optimization strategies to ensure stability and prevent overfitting over a maximum of 200 epochs, utilizing a batch size of 32 and a 15% validation split.

- **Optimizer and Loss Function**: The network weights are optimized using the Adaptive Moment Estimation (Adam) optimizer with an initial learning rate of 1×10^{-4} . The Mean Squared Error (MSE) serves as the primary loss function, with Mean Absolute Error (MAE) tracked as the core evaluation metric.
- **Dynamic Learning Rate (ReduceLROnPlateau)**: To navigate complex local minima during training, the learning rate is dynamically halved if the validation loss fails to improve for 5 consecutive epochs, with a lower bound limit of 1×10^{-6} .
- **Early Stopping**: To maximize generalization on unseen data, an early stopping callback monitors the validation loss. Training is terminated if the validation loss plateaus for 25 consecutive epochs, at which point the model restores the weights from the best-performing epoch.

III. RESULTS

The performance of the proposed hybrid CNN-BiLSTM architecture was evaluated on the 20% hold-out test dataset. Because the architecture was designed to perform continuous regression rather than discrete classification, the primary evaluation metrics center on the absolute and squared deviations between the model's predicted Breaths Per Minute (BPM) and the ground-truth BPM generated by the calibrated RespiPro manikin.

A. Quantitative Error Metrics

The model demonstrated strong predictive capabilities across the continuous respiratory rate spectrum. The overall performance is summarized by Table II:

TABLE II
REGRESSION ERROR ANALYSIS

Metric	Value
Mean Absolute Error (MAE)	2.0564
Root Mean Squared Error (RMSE)	2.9872
Mean Squared Error (MSE)	8.9233
Coefficient of Determination (R^2)	0.8195

The R^2 score of 0.8195 indicates that the extracted spatio-temporal CSI features successfully account for approximately 82% of the variance in the true respiratory rates. The MAE reveals that, on average, the model's prediction deviates from the true respiratory rate by only 2.05 breaths per minute.

B. Clinical Accuracy Thresholds

In continuous vital sign monitoring, exact integer matching is less critical than ensuring the prediction falls within a clinically acceptable margin of error. To assess the practical viability of the model, the predictions were evaluated across expanding tolerance windows (see Table III): These results

TABLE III
TOLERANCE-BASED ACCURACY METRICS

Tolerance-based Accuracy	Percentage (%)
Accuracy within ± 1 BPM	41.66
Accuracy within ± 2 BPM	62.20
Accuracy within ± 3 BPM	75.38
Accuracy within ± 4 BPM	83.90
Accuracy within ± 5 BPM	90.73

show that over 75% of the model's predictions fall within a tight ± 3 BPM window, and over 90% fall within ± 5 BPM, demonstrating high reliability for trend monitoring.

C. Visual Performance Analysis

The training dynamics and prediction distributions are visualized in the following figures:

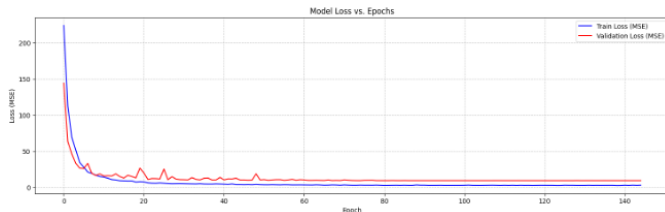


Fig. 5. Loss vs epochs curve

Fig. 5 shows the learning curves indicate a stable convergence. The application of the ReduceLROnPlateau callback effectively mitigated loss oscillations, allowing the validation Mean Squared Error to plateau smoothly without significant

divergence, indicating that the dropout layers successfully prevented overfitting. Fig. 6 presents scatter plot which visually confirms the R^2 score, showing a strong linear correlation tightly clustered along the ideal $y = x$ reference line and Fig. 7 represent a histogram of the residual errors (Predicted - Actual) exhibits a normal, Gaussian-like distribution centered near zero. This indicates that the model does not suffer from significant systematic bias (i.e., it does not systematically over-predict or under-predict the respiratory rate).

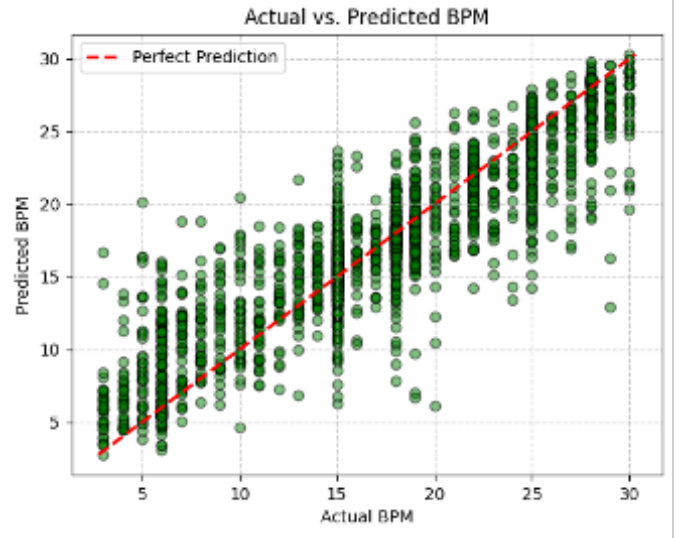


Fig. 6. Actual vs Predicted BPM

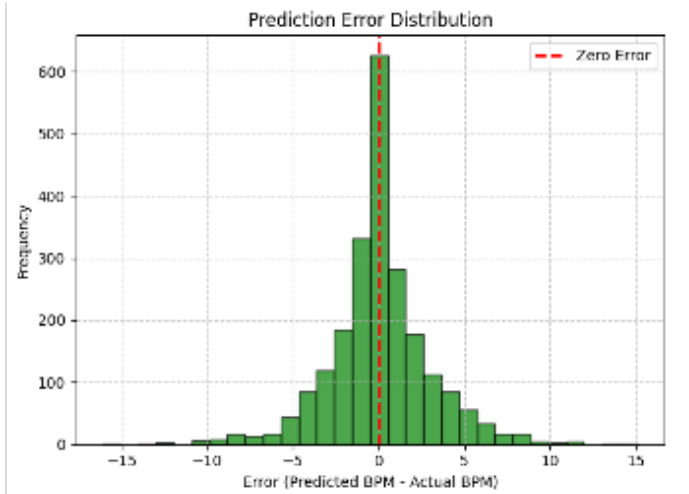


Fig. 7. Prediction Error Distribution

IV. DISCUSSION

The results of this implementation confirm the feasibility and efficacy of using COTS Wi-Fi Channel State Information for contactless, continuous respiratory rate monitoring. By framing the task as a regression problem and deploying a hybrid CNN-BiLSTM architecture, the system successfully

bypassed the computational limitations of traditional mathematical phase-unwrapping techniques while maintaining high accuracy. The Mean Absolute Error of 2.0564 BPM is a highly promising result for a non-invasive RF-based system. In standard clinical environments, a discrepancy of 1 to 2 breaths per minute between a continuous monitor and a manual count is generally considered negligible for tracking overall patient stability. Furthermore, achieving an R^2 of 0.82 demonstrates that the combination of 1D Convolutions (for local high-frequency feature extraction) and Bidirectional LSTMs (for understanding the forward and backward temporal mechanics of a breathing cycle) is a robust topological choice for processing noisy multi-path CSI data. The error distribution (see Fig. 7) provides further confidence in the model's real-world applicability. Because the residuals are normally distributed around zero, the algorithm proves to be unbiased. The 9% of predictions that fall outside the ± 5 BPM threshold likely correspond to edge cases present in the dataset, such as signals subjected to maximum simulated attenuation (e.g., 30 dB of path-loss) or highly abnormal, erratic respiratory patterns where the signal-to-noise ratio is severely degraded. Previous baseline methodologies (such as the original BreatheSmart algorithm) utilized Softmax layers to classify discrete, integer-based breathing rates. While effective, classification restricts the model to predicting only the specific BPMs it was explicitly trained on. By transitioning to a continuous regression model featuring a linear Dense output layer, this implementation allows for the interpolation of fractional breathing rates (e.g., 14.5 BPM). This provides a smoother, more realistic tracking of human respiration as it transitions dynamically between states (e.g., moving from rest to mild physical exertion).

V. CONCLUSION

In conclusion, this study successfully demonstrates the viability of utilizing Commercial Off-The-Shelf (COTS) Wi-Fi CSI for contactless, continuous respiratory rate monitoring. By framing the estimation task as a continuous regression problem and deploying a hybrid CNN-BiLSTM deep learning architecture, the system effectively captured both local high-frequency signal variations and the long-term bidirectional temporal dependencies inherent in the human breathing cycle. Evaluated on a controlled dataset, the model achieved a MAE of 2.0564 BPM and an R^2 score of 0.8195, with over 90% of predictions falling within a clinically acceptable ± 5 BPM margin. These findings validate the potential of data-driven, RF-based sensing methods to bypass the computational overhead of traditional mathematical phase-unwrapping techniques, paving the way for scalable, non-invasive vital sign monitoring in both clinical and residential environments.

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